

Deformation Techniques for Shape Changing Interfaces

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ABSTRACT

Since the introduction of shape changing interfaces, research in the field has contributed a number of taxonomies to classify shape changing interfaces according to different characteristics, including shape, interaction mapping, material, actuation, and information affordances, in an attempt to grasp the diversity of these interfaces in terms of design and information. However, to our knowledge there exists no classifications of input techniques that are used to deform shape changing interfaces through physical interaction. The interaction affordances provided by shape changing interfaces are important for interaction design and interaction mapping. The work presented here aims to analyse how deformable properties in shape changing interfaces are related to deformation techniques, in order to provide a first step towards the development of design guidelines for physical interaction with shape changing interfaces. The results of the study contribute an overview of interaction techniques based on the analysis of a set of 7 models with different deformable properties, each presented in 2D and 3D form.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction techniques; Interaction design.

KEYWORDS

Interaction techniques, Input affordances, Deformable interfaces, Shape changing interfaces, Gesture elicitation

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1 INTRODUCTION

A variety of models and frameworks have been developed to classify shape changing interfaces based on types of shape change [19], interaction mapping [3], design and application [23], resolution [21], material [4, 18], actuation [24], information affordances [17], and control [21]. However, to our knowledge there exists no classification of interaction affordances provided by shape changing interfaces from a physical interaction perspective. Petersen

et al. [18] discuss the definition of affordances in the context of shape changing interfaces from an information perspective based on Gibson's original work on ecological perception and affordances [5]; describing the informative affordances embedded in physical, transformative, and movement characteristics of shape changing interfaces. Despite the inherent haptic qualities of shape changing interfaces, the possibilities for active force feedback in interfaces that utilize physical changes in shape as input and output are rarely discussed. One of the few exceptions is the work by Rasmussen et al. [20] that explores the dynamics between the physical changes in shape and human actions by looking at the levels of control that can be negotiated between the user and the system. However, much more is possible with active haptic force feedback in shape changing interfaces. Force feedback can offer an additional communication layer, as it provides qualities in terms of information, guidance and control [12]. In addition, the shape change and haptic modality exhibit expressive qualities that can complement one another [15]. Further exploration of this perspective of active force feedback in shape changing interfaces requires a comprehensive overview of the interaction techniques that can be performed to deform shape changing interfaces.

The perception of force feedback is highly susceptible to the task that is performed. The amount of force that is applied depends on the grip and the muscles involved in the interaction, which influence the haptic perception [12]. Lederman and Klatzky [10] classified exploratory procedures used to retrieve information about an object using the haptic sense. In this work, we look at the different interaction techniques that can be performed to manually deform shape changing interfaces, starting from a classification of the different types of shape changing interfaces presented in prior work [23]. This classification formed the basis for the development of a preliminary set of interaction techniques used to deform objects or interfaces. We hope that the insights from this work can inspire interaction design with deformable and shape changing interfaces, and support analysis of active haptic force feedback in user interfaces that afford more versatile interaction techniques than conventional interfaces that use buttons, knobs and sliders.

2 RELATED WORK

There is a large diversity in terms of design, technology and application of systems that fit the definition of shape changing interfaces; this is also shown in the considerable effort committed to classify shape changing interfaces according to various characteristics. In the following, we discuss models and frameworks that provide insight into the diversity in deformable properties and input affordances of shape changing and deformable interfaces.

Classifications of Shape Changing Interfaces. Shape changing interfaces are devices that use physical changes of shape or materiality as input and/or output, they are interactive and computationally controlled, self- and/or user-actuated, and have the

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ability to convey information, meaning, or affect [1]. Examples of such interfaces stretch from mobile displays (e.g. [7]) to furniture (e.g. [6]). Over the years, researchers have classified these interfaces from different perspectives. Sturdee and Alexander [23] presented a comprehensive overview of different categorizations methods applied to shape changing interfaces [4, 9, 19, 21, 24] that shows the relation and overlap between different classifications. The model is structured following the interaction framework by Vallgård, a framework that distinguishes between physical, interactive, and temporal aspects of interaction design [27]. The work presented in this paper explores interaction with shape changing interfaces from the perspective of deformation as a result of haptic interaction, and therefore focuses on the classification of shape changing interfaces within the physical dimension [23] as a starting point for the study.

Early on, Coelho and Zigelbaum provided an overview of shape changing materials and their dynamic properties to inform the design of shape changing interfaces from a material perspective [4]. Based on a review of the design space, Rasmussen et al. [19] categorized eight different types of shape change: orientation, form, volume, texture, viscosity, spatiality, adding/subtracting, and permeability. Shortly thereafter, the term shape resolution was introduced by Roudaut et al. [21], providing a metric for the different deformations that can be realized in shape changing interfaces. The user's interpretation of different changes in shape in response to the user's actions was explored by Kwak et al. [9], at different resolutions including deformations in height, volume, orientation and texture. A technical perspective was provided by Taher et al. [24], listing the various actuation techniques used to actuate shape changing interfaces. Recent work by Qamar et al. [18] categorizes self-actuated shape changing interfaces based on their morphing material properties and the 4D printing taxonomy by Nam and Pei provides yet another angle to the material science perspective based on deformations that can be programmed into the material during printing [13].

Interactions with Deformable Interfaces. Surveys of the literature on shape changing interfaces are often focused towards design and development of system output and self-actuation, while systems that only use deformation as means of input are often excluded [3]. To fill this gap, Boem et al. [3] categorized available literature on flexible and malleable deformable interfaces according to their shapes, input sensing, interaction mapping, and input. The materials used for deformable interfaces were categorized as either shape-retaining or non-shape-retaining. The deformable interfaces included in the review supported a variety of hand-based interactions including stretch, bend, twist, squeeze, push, slap, punch, twiddle, roll, and shear.

Materials with different sensing capabilities have been explored to analyse if deformable interfaces can accurately recognize input through deformation (e.g. [14, 28, 30]). Mapping these deformations to actions is a challenging task that has been addressed in various studies (e.g. [8, 11, 25, 26]), mostly in the context of flexible mobile devices and displays. Many of these explorations use flexible or foldable paper or plastic sheets or foam combined with projections to explore the mapping between input and output (e.g. [25, 26]). The shape-retaining properties of foam provide more resistance and allow for more precise input compared to non-shape-retaining flexible materials [22]. Findings from the study by Warren et al.

[31] conducted with a thin flexible prototype suggest that repeating a deformation accurately can be difficult. For interaction with such interfaces, haptic feedback through e.g. material resistance could provide guidance and support for more complex and diverse interactions with flexible interfaces. Instead of looking at the interaction mapping (e.g. [3, 8, 11, 16]) or input recognition (e.g. [14, 31, 32]), the work presented in this paper aims to develop a repertoire of input techniques for the deformation of rigid and flexible interfaces to support further research into active haptics in shape changing interfaces.

3 MATERIALS

A study was set up to analyse the interaction techniques performed to deform different models (Figure 1). The methodology used for the study was inspired by the guessability method often used for gesture elicitation studies [29]. This method was selected to elicit a large number of possible interaction techniques.

The design of the models was based on the classification presented in previous work [23], and focused on the physical characteristics of shape change. The first category is *spatial* and comprises factors like area, volume, height and form. The second category is *orientation* and relates to deformations through curvature, fold and twist. *Resolution* relates to texture, granularity and spatiality. Stretchability, malleability, and viscosity are aspects related to *malleability*. Lastly, permeability and modularity relate to *divisibility*.

Within each category, models were developed with different deformable properties. The material property viscosity was excluded from the analysis, as this property is usually not controlled through direct physical input. The models for each category were presented in rectangular form each in two different versions: 2D and 3D. All 2D models were 8 x 16 cm, and all 3D models were 8 x 16 x 8 cm, this size was selected because it could be handled comfortably with either one or two hands. The rigid models were printed with PLA and flexible filament was used for non-rigid models. The rigid models supported two stable states, while all flexible models return to their original state when there is no force applied. Paper, cloth and textiles interfaces were excluded as this opens up a new design space that has already been explored in prior work (e.g. [3, 11, 26]). There are various ways to design for these individual properties, in particular with characteristics as *resolution* and *divisibility*, which cannot be captured in a small set of models. With the design of this set of models, we aimed for generalizable results that can inform the design of interactions with a large variety of shape changing interfaces by using similar shaped objects that afford a variety of different deformations.

4 STUDY

A within-subject study was designed to explore different interaction techniques with the set of 2D and 3D deformable models (Figure 2). For each model, participants were asked to propose interaction techniques. Four participants (3 male and 1 female) between the age of 28 and 32 ($M=29.3$, $SD=.95$) participated in the study. Three participants were right-handed and one was left-handed. Participants were recruited through the university.

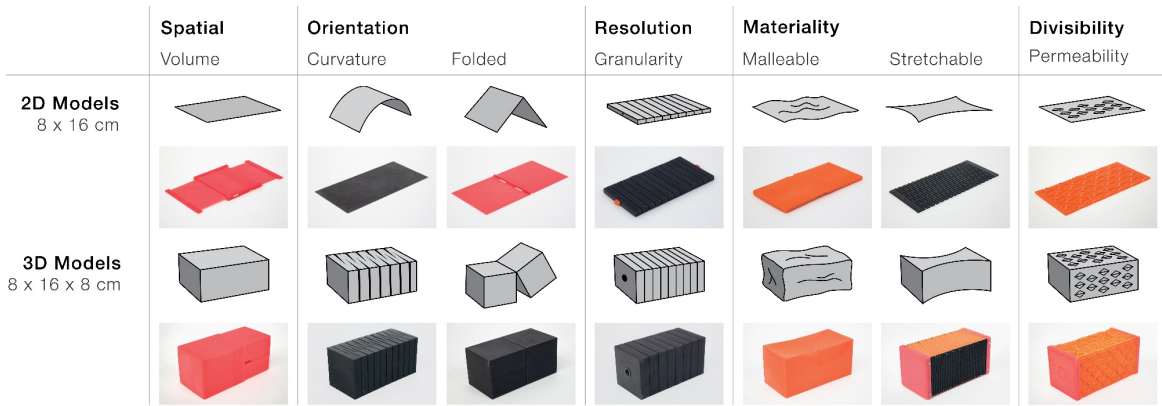


Figure 1: Set of deformable models used for the study. The set is based on classification presented in prior work [24].

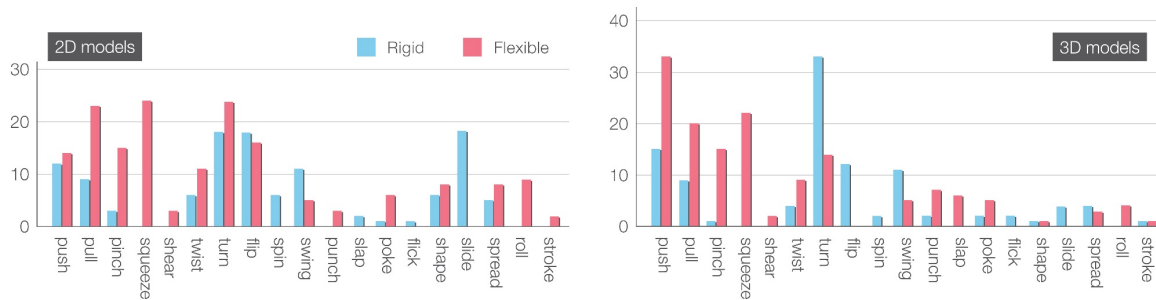


Figure 2: Frequencies of observed interaction techniques for rigid and flexible models in 2D (left) and 3D (right).

4.1 Procedure

Participants were seated behind a desk. A camera was set up to record the interactions, the camera focused on the hands of the participant. The sets of 2D and 3D models were presented individually in counterbalanced order. The models of each set were presented in random order. At the start of each trial, the participant received a model that was designed to afford at least one of the deformations from the list presented in Figure 1. Consequently, the participant was asked to propose any interaction technique they could think of to deform the model corresponding to its deformation type. They were allowed to use either one or both hands, and make use of the table surface as they saw fit. For models with two stable states, interaction techniques were counted for deformations in both ways. For example, opening a foldable model was counted separately from closing the model, as these interactions are not always interchangeable. The study lasted 75 minutes on average.

4.2 Analysis

We collected video recordings of the interactions performed with the models. For each interaction, the start and end frame were saved as well as the timestamps of the frames. The frames were used to create a visual overview of the data that supported the classification of the different interaction methods. The interaction techniques were coded through an iterative process. This resulted in a set of 19 themes to describe the actions performed to deform the models (Figure 2). Further analysis of the interaction techniques focused

on the way the objects were handled, whether the table surface was used, and the grip of the hands.

5 RESULTS

A total of 537 interaction techniques were generated by 4 participants using 14 models, of which 7 were presented in 2D form (288 interaction techniques) and 7 in 3D form (249 interaction techniques). The following section discusses the manner in which participants handled the models, and the interaction techniques used to deform rigid and flexible models.

5.1 Object Handling and Surface Usage

The video recordings showed that the way participants handled the objects varied. Four different types of object handling were identified: single-handed interactions with a subset of the fingers or thumb, single-handed interaction with the whole hand, symmetric bimanual interactions, and asymmetric bimanual interactions (Figure 3).

5.1.1 Single-handed Interactions. In total, 172 single-handed interactions were recorded, 69% of these interactions were performed with the whole hand, while the remaining 31% were performed with a subset of the fingers or the thumb. Single-handed interaction techniques were often performed with the aid of the table surface (56.4%). We observed that 3D models that were deformed with single-handed interactions with only a subset of the fingers or

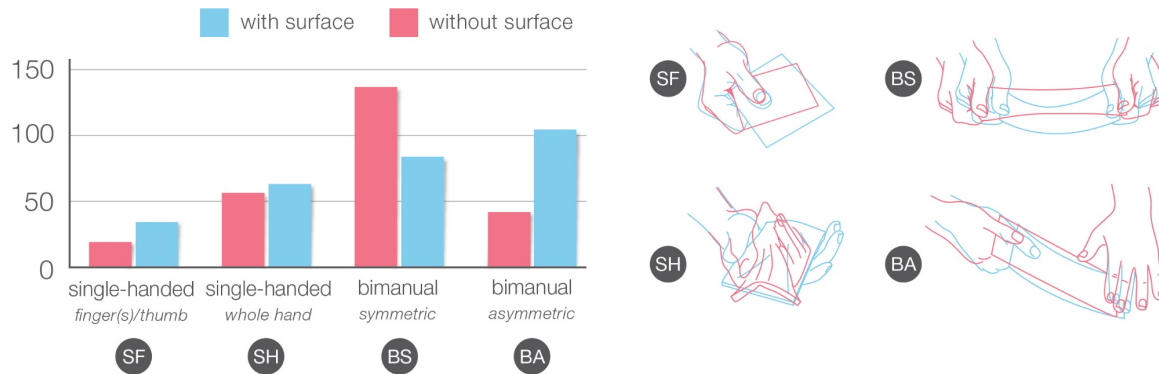


Figure 3: Frequencies of the observed single-handed and bimanual interactions with and without use of the table surface.

the thumb required the use of a surface. Single-handed interactions with the thumb without the aid of the table surface were mainly observed with 2D rigid models as these models could be clamped between the fingers and the palm of the hand more easily when solely the thumb was engaged in the interaction.

5.1.2 Bimanual Interactions. The majority of the interactions (365) were performed with two hands. The most frequently used technique was symmetric bimanual interaction (60%), in which the same interaction technique was observed in both hands either identical or opposite. The symmetric bimanual interaction technique was the only technique that was observed more frequently without support from a surface as shown by the result of a Chi-Square test $\chi^2(1) = 33.34, p < .01$. For the remaining 40%, participants used asymmetrical bimanual interactions, in which the non-dominant hand was observed supporting the model. The opposite pattern could be observed here, with significantly less interactions performed without support from the table surface: $\chi^2(1) = 27.09, p < .01$. When the surface was used for asymmetrical bimanual interaction, the non-dominant hand would keep the model in place often with a radial grasp, flat hand, fingertips, or pinch grip. The grasp technique depended on the characteristics of the model and type of interaction that was performed.

5.2 Interaction Techniques

Based on the analysis of the recorded interactions, 19 interaction techniques were identified (Figure 4). As expected, different types of models afforded different types of interactions that corresponded with the model properties. The frequencies with which these techniques were used to perform deformations in rigid and flexible models are presented in Figure 2. Here, we discuss the interaction techniques that were typically used for rigid and flexible models. Followed by interaction techniques that were universally applicable.

5.2.1 Typical Interaction Techniques for Rigid Models. Some of the interaction techniques performed to change the orientation or volume were associated with rigid models. Typical interaction techniques to apply a change in the orientation, which were usually observed in rigid models, were flipping, spinning, and flicking. Additionally, sliding methods were observed for changes in volume of rigid models.

Slide. The sliding interactions could be performed with finger and or thumb extension and flexion. By positioning multiple fingers on a surface, friction is created between the hand and the object. Sliding was one of the few methods that could be performed single-handed with the thumb, without support from a surface.

Flip, Spin and Flick. These three interactions were mostly performed to alter the orientation of rigid models. Flipping seems most common in 2D foldable models, while spinning was only possible with the granularity model, which was the only model that supported full rotation. All three interaction techniques can be used to initiate momentum, for flipping and flicking this is achieved with finger and/or thumb flexion or extension. For spinning, momentum is created from the rotary motion of the wrist or shoulder. These deformation methods are quick, while at the same time the output of the resulting deformation is difficult to control.

5.2.2 Typical Interaction Techniques for Flexible Models. Flexible models supported different types of interaction that were not observed in interactions with rigid models. These interactions include methods to apply pressure such as pinching, squeezing, and punching. Furthermore, flipping interactions were also observed in 2D curvature models as these could be ‘flipped’ under tension and would return to their original state after pressure was released. In these models, flipping is achieved by a slight change in orientation of the grip, which was observed single-handed or symmetric bimanual.

Pinch, Squeeze and Punch. The malleable model in particular invited for these types of interactions. Commonly used pinches were lateral, pulp, and five-finger pinch. Squeezing interactions varied from subtle squeezes with power grasps to crumbling. Lastly, punches were performed with clenched fist exclusively on malleable models.

5.2.3 Universal Interaction Techniques. The following interaction techniques were observed in flexible and rigid models.

Push, Pull and Shear. Pushing was a common method to apply pressure with a spherical grasp, flat hand(s), extended finger(s), thumb(s), or with a pinch grasp. Whereas, pulling is a common technique often performed with spherical or pinch grasp, but can also be performed with a hook and snap grasp, which was observed in interaction with permeable models. For flexible models, changes

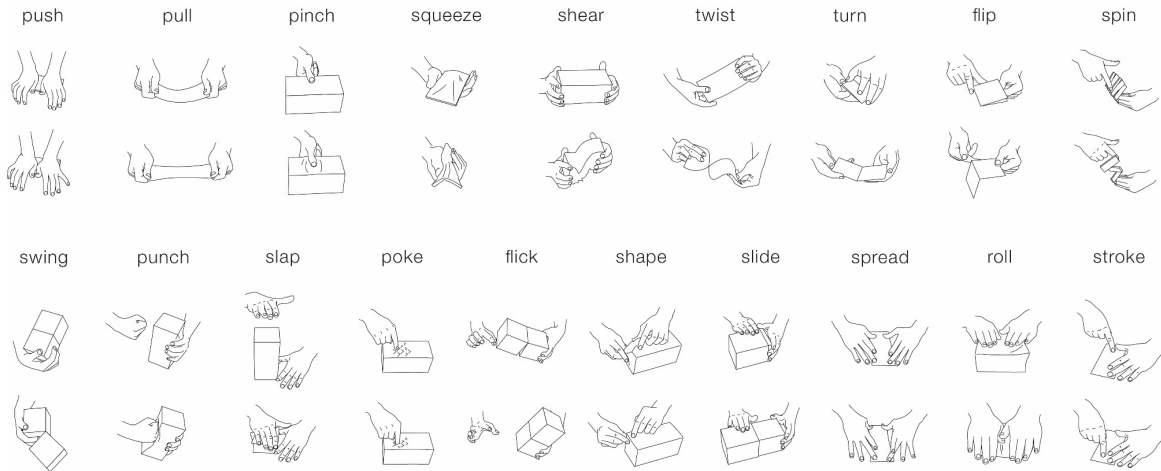


Figure 4: Examples for each of the 19 interaction techniques observed in the study.

in the directional movement of push and pull actions were used to create changes in the orientation of the model (curvature or shear).

Twist, Turn and Swing. Orientation changes were often generated with twist and turn interactions. Twisting was symmetric bimanual as it requires movement of both hands rotating in opposite direction, while turning was mostly performed with asymmetrical bimanual object handling using the non-dominant hand to keep the model in place. Swing interactions on the other hand were often performed single-handed, using the momentum of the arm to inflict a deformation in the model's volume or orientation. For all these interactions, the grasps used by participants were similar depending on the characteristics of the model. Observed grasps include spherical grasp, extension grasp, and lateral pinch.

Roll, Spread and Stroke. Rolling interactions were only observed in flexible models, due to the rectangular shape used for the models. Therefore, rolling is listed as an interaction technique that could additionally be used in rigid models when this is supported by its design. Rolling interactions were often performed with a flat hand or with extended finger(s), using the contact surface between the hand and material similar to sliding techniques. Spread interactions were initiated by extension of the thumb and/or a subset of fingers, to stretch flexible materials or change the orientation or volume of rigid models. Stroke interactions were observed in flexible and modular (granularity) models, performed with extension of a subset of fingers. Roll, spread and stroke interactions can be performed single-handed with the aid of a surface or bimanual.

Poke and Slap. These interaction techniques are used to apply a deformation as a result of a collision with the model. This can be subtle with an extended finger when poking, or with a flat hand when slapping. These interaction techniques are versatile, but were most frequently observed in permeable and malleable models.

Shape. For interaction with the 2D models for granularity (rigid) and curvature (flexible) a different type of interaction technique was observed in which the participant used the shape of their hands to align the individual elements of the model (granularity) or shape the curvature or form of flexible models. This interaction technique might become more relevant for flexible and modular interfaces.

6 DISCUSSION AND CONCLUSIONS

Shape changing interfaces can be more versatile in terms of interaction affordance compared to conventional interfaces. With conventional interfaces that utilize force feedback, the physical design often clearly directs the way users interact with the device through sliders, knobs, levers and the like. With these controls, there is still room for differences in grasp and body posture that affect the haptic experience, however, this is often more restricted compared to interaction affordances of shape changing interfaces. Shape changing interfaces have the ability to take on a variety of different shapes depending on their deformable properties. These can (but do not have to) provide more diversity in physical appearance, affordances, and the way the interface can be handled, as there is often no longer a single point for interaction.

Inform Force Feedback Design and Evaluation of Shape Changing Interfaces. When such interfaces utilize force feedback for guidance, communication, or feedback purposes, it becomes important to evaluate the feedback for different interaction techniques, and adapt the design when needed to create a holistic haptic experience regardless of the manner in which the object is handled. This work attempts to provide an overview for interaction techniques that can be used to aid the evaluation of haptic interaction with shape changing interfaces and for the development of haptic design guidelines for more consistent and reliable haptic feedback in shape changing interfaces that utilize active force feedback for guidance, feedback, or communication.

Force feedback could not only support interactions with rigid shape changing interfaces. It also has the potential to improve interactions with flexible interfaces considering the challenges that were observed with single-handed object handling of flexible interfaces and the difficulties with replicating input [31], local adjustable resistance could be a way to guide interactions and allow for more precise and diverse input methods in flexible and malleable interfaces.

Inspire Interaction Mapping with 2D and 3D Interfaces with Deformable Properties. In addition, the work presented provides an overview as well as analysis of the object handling with

interfaces that exhibit different deformable properties. This can inspire the design of deformable characteristics in shape changing interfaces and mappings between interaction techniques and the system's behavior or functionality.

Another interesting potential of interfaces that provide more diversity in physical appearance, affordances, and object handling is the possibility to detect expressive user input based on the way the user interacts with the interface. Future work could explore how different interaction techniques and their parameters relate to the affective state of the user (e.g. [2]).

6.1 Limitations and Future Work

Several limitations to this work need to be addressed. First, the grasp technique is an important aspect that influences the haptic experience of an interaction. In this study, the models afforded a variety of grasping techniques for hand-based interaction. However, these techniques were restricted to the design of the models. For example, the use of handles or levers affect the affordances of the interaction as well as the techniques that can be employed.

Second, the initial set presented here shows how different interaction techniques relate to different deformable qualities of shape changing interfaces based on a preliminary study with a small sample size. Therefore, these insights serve as a starting point rather than a complete overview of interaction techniques with shape changing interfaces. A set that could be expanded on in future research.

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